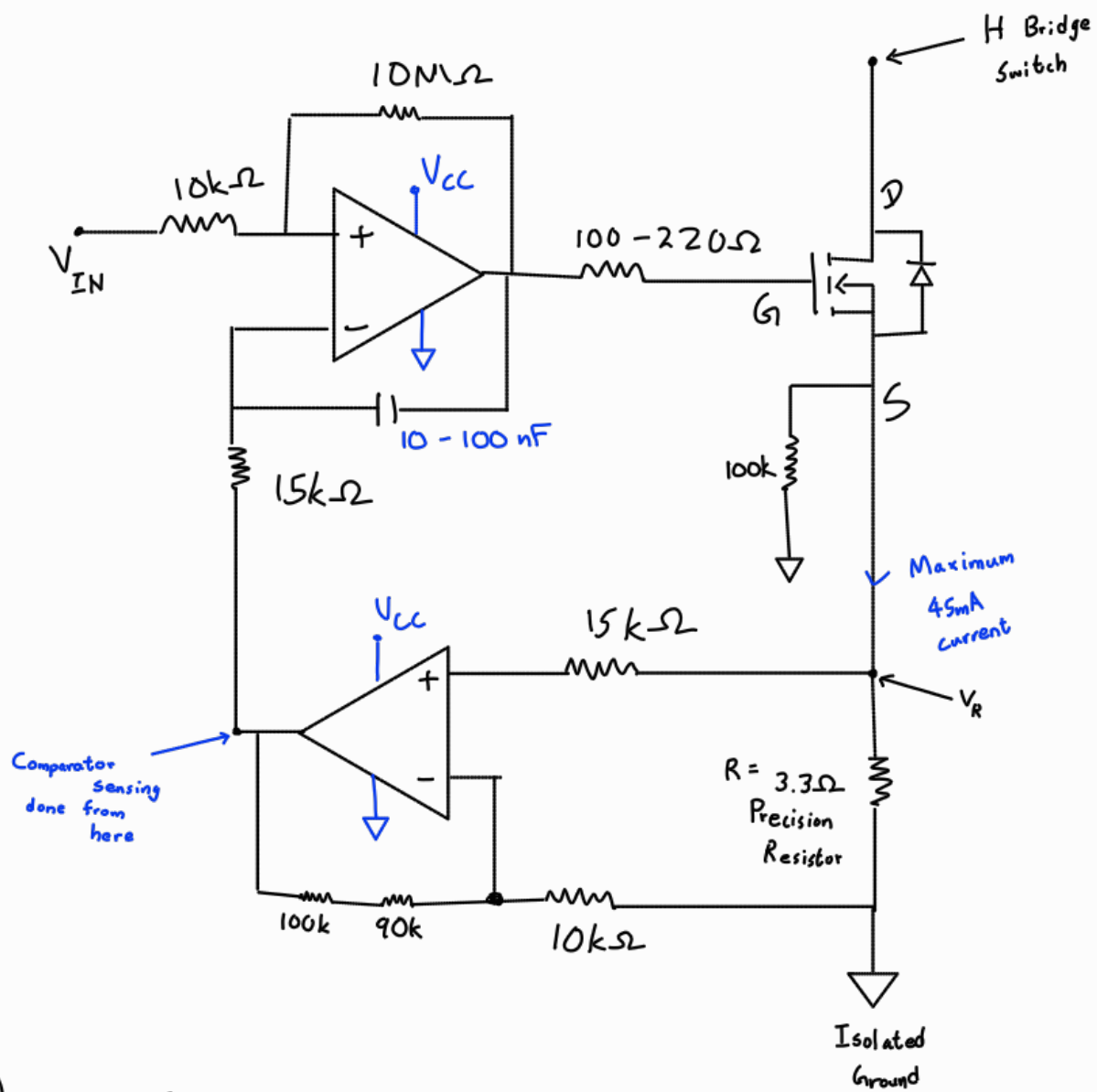




$$(V_{in})_{\max} = 3V$$

$$(V_R)_{\max} = \frac{1}{20} (V_{in})_{\max}$$

$$I_{\max} \times R = \frac{1}{20} \times (V_{in})_{\max}$$

$$\therefore R = \frac{3}{20 \times 45 \times 10^{-3}}$$

$$R = 3.3\Omega$$